Repeated Contention Scheduling: A Novel Resource Allocation Algorithm Toward 6G Vehicular Networks

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***Abstract*—Efficient decentralized resource allocation remains a fundamental challenge in NR-V2X sidelink communications, where conventional Semi-Persistent Scheduling (SPS) and Dynamic Scheduling (DS) suffer from persistent collisions and limited adaptability under dynamic and dense conditions. This paper proposes Repeated Contention Scheduling (RCS), a novel resource allocation algorithm based on multi-round, feedback-driven contention that eliminates long-term reservations and enables fully distributed operation. Simulations demonstrate that RCS outperforms SPS and DS in terms of success probability, collision and loss reduction, and timeliness metrics such as Packet Inter-Reception Delay and Age of Information, particularly under high load. The practical feasibility of the approach is validated through an SDR-based experimental testbed, which confirms robust operation under realistic hardware impairments and closely matches theoretical and simulation results. These findings establish RCS as a viable and scalable solution for resource allocation in 5G NR sidelink and a promising candidate for future 6G vehicular communication systems.**

***Index Terms*—5G, 6G, V2X, Semi-Persistent Scheduling, Sidelink, Age of Information, Dynamic Scheduling, Persistence, Vehicular Networks, Repeated Contention, Vehicular Communication, Resource Allocation.**

I. Introduction

Intelligent Transportation Systems (ITS) constitute a key component of next-generation transportation systems, enabling improved safety, traffic efficiency, and automation through pervasive connectivity. Vehicular networks provide the communication backbone of ITS, supporting real-time information exchange among vehicles, infrastructure, and vulnerable road users. In this context, 5G New Radio (NR)-Vehicle-to-Everything (V2X) has emerged as a technological solution offering high-reliability, low-latency communication tailored to advanced vehicular applications [1]. A central feature of NR-V2X is sidelink communication, which enables direct and decentralized interactions in both in-coverage (Mode 1) and out-of-coverage (Mode 2) scenarios [1, 2].

Efficient sidelink resource allocation remains a fundamental challenge in dynamic vehicular environments characterized by high mobility, heterogeneous traffic density, and rapidly varying channel conditions. The 5G NR-V2X sidelink framework defines two operational modes. In Mode 1, the gNB performs centralized scheduling through dynamic grants for per-transmission allocation and configured grants for semi-persistent reservations, supporting both periodic and aperiodic traffic under network coverage. In Mode 2, user equipments autonomously select resources based on sensing mechanisms. Two standardized allocation schemes are defined for this mode: Semi-Persistent Scheduling (SPS) and Dynamic Scheduling (DS) [3, 4].

SPS adopts a reservation-based strategy in which resources are selected and retained over multiple transmission intervals according to a predefined Resource Reservation Interval (RRI), controlled by parameters such as the Reselection Counter (RC) and persistence probability P_{pers}. In contrast, DS follows a non-persistent approach, where resources are selected independently for each packet, making it more suitable for aperiodic traffic [5]. Although both schemes are standardized, no explicit guideline is provided for their selection, and their effectiveness depends on traffic characteristics and network conditions.

SPS is widely adopted due to its efficiency for periodic traffic and reduced control overhead in decentralized operation. However, its persistence-based design introduces inherent limitations. Repeated reuse of reserved resources increases the probability of persistent packet collisions when multiple vehicles select overlapping resources, particularly in the presence of hidden terminals and sensing inaccuracies [6]. These collisions may persist over consecutive transmissions [6, 7, 8], leading to inefficient spectrum utilization [5] and degraded quality of service under high mobility or network congestion [9].

To mitigate these issues, numerous enhancements have been proposed. At the protocol level, 3GPP introduces mechanisms such as short-term sensing [10], resource re-evaluation [11, 12, 13], pre-emption [14], and inter-UE coordination [15], along with feedback-based reliability features including Hybrid Automatic Repeat Request (HARQ) and Physical sidelink Feedback Channel (PSFCH) [16, 17, 18]. From an algorithmic perspective, optimization-based approaches, including convex optimization, game theory, and graph-based formulations, have been applied to improve resource selection [19]. In parallel, machine learning approaches, particularly reinforcement learning [20, 21, 22, 23] and multi-agent systems [24, 25], enable adaptive and context-aware decision-making [26, 27, 28, 29, 30, 31, 32] in decentralized environments. Additional directions include the integration of Non-Orthogonal Multiple Access (NOMA) for enhanced spectral efficiency and context-aware strategies such as RSU-assisted or position-based allocation [33, 34, 35, 36].

Despite these advances, most approaches remain evolutionary and retain the semi-persistent allocation paradigm, thereby remaining susceptible to recurring collisions. Consequently, robust distributed resource allocation remains an open problem. Survey and tutorial studies on resource allocation in 5G NR-V2X and beyond [37, 38, 39, 40] confirm that recent progress has largely focused on incremental refinements of SPS/DS-based mechanisms rather than fundamentally new designs. As a result, persistent collisions caused by hidden terminals, sensing inaccuracies, and limited coordination remain unresolved. Moreover, non-persistent allocation strategies have received comparatively limited attention, particularly those

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incorporating effective feedback and coordination without relying on long-term reservations. This highlights that it is worth investigating alternative paradigms that overcome the inherent limitations of reservation-based schemes.

Motivated by these limitations, this paper proposes a new sidelink resource allocation approach, termed Repeated Contention Scheduling (RCS), for NR-V2X and future 6G-V2X systems. Unlike SPS, the proposed method *is not based on semi-persistent reservations*. Instead, it employs repeated contention rounds [41, 42, 43], where vehicles contend for resources for each message they have to send. Contention exploits sub-carrier signaling with simultaneous transmission and reception. The feasibility of this key function is proved via an experimental test-bed. This design eliminates long-term reservations and replaces them with a dynamically re-evaluated contention process. RCS is fully compatible with 5G NR-V2X sidelink while improving reliability and timeliness in dense scenarios, as highlighted by numerical results. RCS provides a scalable solution for current vehicular networks and a foundation for future intelligent resource allocation mechanisms in 6G vehicular systems.

Main contribution of the paper:

- We propose a novel resource allocation algorithm, termed RCS, based on a repeated contention mechanism for NR-V2X sidelink communication. Unlike conventional DS and SPS schemes, the proposed approach eliminates semi-persistent reservations and instead relies on feedback-driven contention rounds, enabling dynamic and collision-resilient resource selection.
- We demonstrate through extensive simulations that RCS outperforms traditional DS and SPS in terms of reliability, collision mitigation, information timeliness, particularly in dense vehicular scenarios.
- We validate the practical applicability of RCS through a real-world experimental setup, providing a proof-of-concept implementation that confirms its feasibility and effectiveness under realistic deployment conditions.

The rest of this paper is organized as follows. Section II presents the proposed RCS algorithm, detailing its repeated contention mechanism and design principles. Section III describes the simulation framework, including system assumptions, parameter configuration, and evaluation metrics relevant to reliability and timeliness. Also, this section reports the simulation results and provides a comparative performance analysis against baseline schemes. Section IV presents the experimental proof-of-concept implementation and discusses practical considerations and observed performance in a real-world setup, as illustrated in the prototype system. Finally, Section V summarizes the main findings and outlines directions for future research.

II. Description of RCS

In this section, we describe the system model and the configuration of 5G NR multiple access to support RCS. Then, the RCS algorithm is presented in Section II-B. Then we provide an analysis of RCS (Section II-C).

A. System model

Let us consider n nodes, each one generating update messages. We assume a message fits into a single Sub-Channel (SC). The average generation time, i.e., the mean time elapsing for a node to generate a new update message, is denoted with T_g.

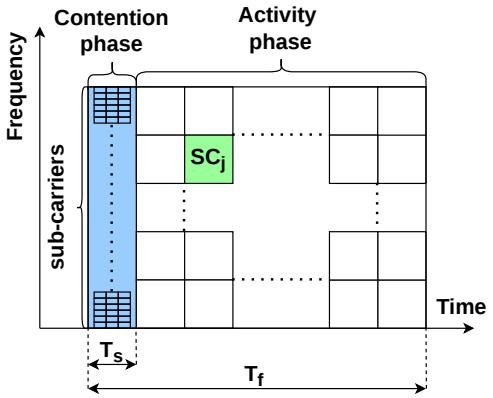


Figure 1. Configuration of the channel using RCS.

Table I
Used symbols and corresponding definitions.

Parameter	Definition
n	Number of nodes
T_g	Message generation interval
T_s	Slot duration
T_f	Frame duration, $T_f = n_f T_s$
n_f	Number of slots per frame
n_{SC}	Number of SCs per slot
K	Number of SCs per frame, $K = (n_f - 1)n_{\text{SC}}$
a	System load, given by $a = (n/T_g)/(K/T_f)$
M	Total number of sub-carriers used during contention
J	Number of contention groups (divisor of K)
w	Number of SCs associated with one contention group
m	Sub-carriers per group, $m = \lfloor M/J \rfloor$
δ	Duration of one contention round
T_{DB}	Packet delay budget

The multiple access structure is shown in Figure 1. It consists of frames, each frame comprising n_f 5G NR sub-frames, hence lasting $T_f = n_f T_s$, where T_s is the sub-frame time (referred to as slot time in the following). Each frame comprises a *contention time*, lasting one or more time slots, and SCs in all remaining time slots, where SC definition and structure are as in sidelink. The number of slot times devoted to contention time depends on the selected numerology. In the case of $\mu = 0$ (reference numerology), it is assumed that contention time reduces to its minimum, one time slot. Hence, the remaining $n_f - 1$ slots of a frame are used to carry SCs. If n_{SC} SCs are configured in each time slot, the overall number of SCs in one frame is given by $K = (n_f - 1)n_{\text{SC}}$. It is then possible to define a load factor a as follows:

$$a = \frac{n\,T_f}{K\,T_g} \tag{1}$$

Note that, if $a > 1$, the system is structurally overloaded, i.e., there are not enough SCs for all generated messages.

A delay budget is associated to each message. If the time elapsing since when the message was generated exceeds the delay budget and the message has not been transmitted yet, the message is discarded, without being transmitted. In the following, we assume that the delay budget is the same for all messages, denoted with T_{DB}. To guarantee that at least one contention is possible irrespective of the generation time of the message, it must be $T_{\text{DB}} \geq 2T_f$. On the other hand, there is no point in setting $T_{\text{DB}} > T_g$, given that a new update message is generated every time period T_g. Hence, we assume $2T_f \leq T_{\text{DB}} \leq T_g$ (holding for $T_g > 2T_f$).

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B. RCS algorithm

Let us follow the operation of a backlogged node, first introducing the baseline contention procedure (single-win). Generalizations to multiple-win are then introduced. Used notation is listed in Table I.

1) Basic RCS algorithm: As soon as a new message is available, say at time t_0, the backlogged node waits until the first occurrence of the contention time after t_0. Then it starts the contention procedure for that message. Since there is one contention time per frame, the node must wait at most for a time $T_f - T_s$, where we account for a contention time equal to one slot time. A set of sub-carriers is designated out of all sub-carriers in the sidelink channel.[1] Let M be the overall number of sub-carriers used in the contention procedure. Since there are K SCs to be assigned in one frame, sub-carriers are organized into K subsets, each comprising m sub-carriers. Sub-carrier subset j is associated to the j-th SC of the frame, $j = 1, \ldots, K$.[2] A backlogged node selects one sub-carrier subset at random and contends for the SC associated with that subset.

The contention time is organized into r consecutive rounds. At the beginning of the first round, the node selects at random one out of the m sub-carriers of the chosen subset. Let $f_1, f_2, \ldots, f_m$ be the frequencies of the m sub-carriers, labeled in increasing order, i.e., so that $f_i < f_j$ if $i < j$. Let f_x be the frequency picked at random by the considered node. Each contending node transmits its selected sub-carrier for the whole round and *simultaneously* the node listens to the frequency band comprising the selected sub-carrier group, to identify sub-carriers transmitted by other nodes. If the considered node detects a frequency $f_y < f_x$, then the node deems itself as losing the contention and leaves the current contention. It will take part in the next contention (after one frame time), if the packet delay budget allows this additional delay. If instead the considered node does not detect any frequency less than its own selected frequency f_x, then it will deem itself as a winner, and it will move to the next round. This contention procedure is repeated the same in every round, each time independently selecting a sub-carrier at random from the considered subset. Nodes surviving up to the last round included (the r-th round) are the final winners. Winners of sub-carrier subset j are entitled to use SC j in the frame where they have won the contention. A collision will occur if and only if there is more than one winner. The analysis in Section II-C will show that the probability of collision decays exponentially fast with the number of rounds.

After having won the contention and having transmitted in the SC associated with the selected sub-carrier subset, the node goes back to idle if it has no other pending message. Otherwise, it starts the whole procedure all over again for the next scheduled message.

2) Generalized RCS algorithm: Splitting sub-carriers into K groups and letting contending nodes joining groups at random may lead to some of the groups being void, i.e., with no contending node. Then, the associated SC would go unused in that frame. To mitigate this potential waste of resource, we generalize the scheme of the previous sub-section as follows. Let us define $J \leq K$ resource groups, where J is chosen as a divisor of K. Each resource group comprises $w = K/J$ distinct SCs of a frame. The basic procedure defined in the previous subsection is the special case where $J = K$ and hence $w = 1$. Since multiple SCs are associated with each group, the contention in a group defines multiple winners. More in depth, exactly w winners should be ideally identified. The number of actual winners in a given contention may fall short of w, because too few nodes joined that group. It may also exceed w, in case multiple nodes select the same winning sub-carriers (then, a collision event will occur for at least one of the SCs belonging to the group).

To define the generalized procedure, the whole set of M sub-carriers used for contention is split into J subsets, each comprising $m = \lfloor M/J \rfloor$ sub-carriers. Assume a contending node selects a sub-carrier with frequency f_x. The contending node wins if it detects *less* than w sub-carriers with frequency $f_y < f_x$. On the contrary, if at least w sub-carriers with frequency $f_y < f_x$ are detected, the node leaves the contention. This generalized contention is referred to as "multi-win".

3) Implementation of the RCS algorithm: The critical points to implement the RCS algorithm in 5G NR and towards 6G sidelink are as follows.

- Simultaneous transmission and reception of sub-carriers must be feasible.
- Nodes must be synchronized so that boundaries of contention rounds are aligned.

As for the first point, note that RCS does not require implementing a true full-duplex radio. The requirement to implement RCS is only that a node be able to transmit a tone at frequency f_x and *at the same time* detect tones in a frequency band between f_1 and f_{x-1}, where f_x is the tone frequency selected by the node. The feasibility of this mild form of duplex is proved in experiments described in Section IV.

As for the second point, the duration δ of a contention round should be one or a few symbol times (with numerology $\mu = 0$, a sub-frame lasts 1 ms and contains 14 OFDM symbols, hence one symbol lasts $\approx 71\,\mu$s). We will see in Section IV that a round time of 3 symbol times is enough to guarantee reliable simultaneous sub-carrier transmission and detection. Tight synchronization among nodes operating in sidelink should be guaranteed anyway, independently of RCS, to guarantee correct detection of SC control information and carried data.

C. Analysis of RCS collision probability

A model is presented to evaluate the collision probability of the basic RCS contention procedure [43]. The model holds under the assumption that each contending node detects sub-carriers transmitted by all other contending nodes.

Let n be the number of nodes backlogged at the beginning of the contention phase. Let q_i denote the probability that a node picks frequency f_i, $i = 1, \ldots, m$. Let also $G_i = \sum_{j=i}^{m} q_j$ be the corresponding Complementary Cumulative Distribution Function (CCDF).

We define a Discrete Time Markov Chain (DTMC) $X(t)$, where $X(t)$ denotes the number of nodes contending in round $t = 1, \ldots, r$. $X(0) = n$ is the initial number of contending nodes. The DTMC evolves over the state space $\{1, \ldots, n\}$. The one-step transition probability matrix of the DTMC is denoted with $\mathbf{P}$, with entries given by:

$$P_{k,h} = \begin{cases} \sum_{i=1}^{m-1} \binom{k}{h} q_i^h G_{i+1}^{k-h} & h = 1, \ldots, k-1 \\ \sum_{i=1}^{m} q_i^k & h = k, \\ 0 & h = k+1, \ldots, n, \end{cases} \quad (2)$$

[1] We will see in Section IV that reliable detection of sub-carriers requires a frequency separation between sub-carriers used in the contention procedure.

[2] SCs belonging to one frame can always be numbered in a non-ambiguous order.

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for $k = 1, \ldots, n$.

Let $\mathbf{p}(t)$ denote the state probability vector at time t, the k-th component of which is $p_k(t) = \mathcal{P}(X(t) = k)$, $k = 1, \ldots, n$. At the outset of the contention we have $p_n(0) = 1$ and $p_k(0) = 0$, $k = 1, \ldots, n-1$. In the following, we choose a uniform probability distribution for sub-carrier selection in all rounds, i.e., we let $q_i = 1/m$, $i = 1, \ldots, m$, for every round. The corresponding matrix having entries as given in Equation (2) is denoted with $\mathbf{P}_u$, to emphasize that a uniform probability distribution has been adopted for sub-carrier selection. Therefore, the probability distribution of the state of the DTMC at time t is $\mathbf{p}(t) = \mathbf{p}(0)\mathbf{P}_u^t$, for $t = 1, \ldots, r$. The probability of a successful contention, i.e., that a single node wins the last round, is $P_{\text{succ}} = \mathcal{P}(X(r) = 1) = p_1(r)$. Correspondingly, the collision probability is $P_{\text{coll}} = \mathcal{P}(X(r) > 1) = 1 - p_1(r)$. Let $\mathbf{Q}_u$ denote the square matrix obtained by taking the last $n-1$ rows and columns of $\mathbf{P}_u$. The collision probability can be written as

$$P_{\text{coll}} = \mathbf{y}\, \mathbf{Q}_u^r\, \mathbf{e} \tag{3}$$

where $\mathbf{y}$ is a row vector containing the last $n-1$ components of $\mathbf{p}(0)$, i.e., $y_{n-1} = 1$, $y_k = 0$, $k = 1, \ldots, n-2$, and $\mathbf{e}$ is a column vector of 1's. The matrix $\mathbf{Q}_u$ is lower triangular, with diagonal elements given by $P_{u,kk}$ in Equation (2) for $k = 2, \ldots, n$. Hence, its dominant eigenvalue is $\eta = Q_{u,11} = P_{u,22} = \sum_{i=1}^{m} q_i^2 = 1/m$, where we have used the fact that $q_i = 1/m$, $\forall i$. Since $\mathbf{Q}_u$ is also a non-negative matrix, the right eigenvector $\mathbf{u}$ associated to η is positive. It is possible to find the closed form of $\mathbf{u}$, namely $\mathbf{u}^T = [2\ 3\ \ldots\ n]/2$, where the superscript ^{T} denotes transposition. Since it is $\mathbf{e}^T = [1\ 1 \ldots 1] \leq [2\ 3 \ldots n]/2 = \mathbf{u}^T$ and all vectors and matrices are non-negative, we have:

$$P_{\text{coll}} = \mathbf{y}\mathbf{Q}_u^r\mathbf{e} \leq \mathbf{y}\mathbf{Q}_u^r\mathbf{u} = \frac{1}{m^r}\mathbf{y}\mathbf{u} = \frac{n}{2m^r} \tag{4}$$

In the end, we find the following upper bound on the collision probability for any value of the number of contending nodes, n:

$$P_{\text{coll}} \leq \min\left\{1, \frac{n}{2m^r}\right\} \tag{5}$$

The key feature of the upper bound is that it reveals that the collision probability decays exponentially as the number of rounds r grows. It also highlights the role of the number m of sub-carriers used in the contention subset. The analysis developed in this Section and the resulting upper bound hold as long as nodes can hear each other and sub-carrier detection is error-free.

III. Simulations results

This section presents the simulation-based evaluation of the proposed RCS scheme. Section III-A defines the metrics used to assess both reliability and timeliness of packet delivery. The simulation setup is described in Section III-B. Section III-C analyzes the impact of the number of contention groups on reliability and timeliness over a wide range of load conditions. Finally, Section III-D compares the performance of RCS with the standardized sidelink multiple access schemes SPS and DS.

A. Key performance metrics for simulation of RCS

It is assumed that all nodes can hear one another. As a consequence, sub-carrier signaling works as described in Section II-C. An all-or-nothing model is used for the communication channel. Then, packet reception is successful except in the case of collision (more than one node transmitting on the same SC). Hence, under RCS possible outcomes of packet P generated by node A are as follows: (i) node A is the unique winner of an SC resource for packet P; then P will be received successfully by all other nodes; (ii) multiple nodes win the same SC as A; then packet P incurs a collision event that disrupts its reception; (iii) packet P is dropped (after having lost all available contentions), because it exceeds its delay budget. The considered reliability metrics are related to outcomes of a packet: Packet Reception Ratio (PRR), collision probability ($P_{\text{coll}}(sim)$), and packet drop probability (P_{drop}), with $PRR + P_{\text{coll}} + P_{\text{drop}} = 1$. The PRR is defined as the fraction of successfully delivered packets:

$$\text{PRR} = \frac{N_{\text{singlewin}}}{N_{\text{total}}} \tag{6}$$

The collision probability is defined as:

$$P_{\text{coll}}(sim) = \frac{N_{\text{multiplewin}}}{N_{\text{total}}} \tag{7}$$

The packet drop probability accounts for delay constraints and is defined as:

$$P_{\text{drop}} = \frac{N_{\text{drop}}}{N_{\text{total}}}, \tag{8}$$

where a packet is discarded if its delay exceeds T_{DB}. A packet gets delayed when it loses a contention, and it has to go to the next contention. Since each packet is associated with a deadline, if the elapsed time from generation exceeds this limit, the packet is considered obsolete and is discarded, giving up on running more contention.

The timeliness of the system is assessed through Packet Inter-Reception Delay (PIR) [44], mean Age of Information (AoI), and the AoI violation probability. The PIR [44] quantifies the time interval between two consecutive successful receptions of packets belonging to the same application flow, thus capturing the effective update timing at the receiver. Consider the local dynamic map maintained at node j, which aggregates information received from neighboring nodes. Let $t_{ij}(g)$ denote the reception time at node j of the g-th successfully received packet from node i, with $g \geq 2$. AoI at node j for updates from node i is defined as

$$A_{ij}(t) = t - t_{ij}(g-1), \quad t \in [t_{ij}(g-1),\, t_{ij}(g)) \tag{9}$$

The peak AoI, which coincides with PIR, corresponding to the maximum age just before a new reception, is therefore

$$Y_{ij}(g) = t_{ij}(g) - t_{ij}(g-1) \tag{10}$$

Let z_{ij} denote the number of successfully received updates from node i to node j. The average PIR at node j, accounting for all incoming flows (i, j), is computed as a weighted average:

$$\text{E}[\text{PIR}_j] = \sum_{i \in \mathcal{N}_j} \frac{\sum_{g=1}^{z_{ij}} Y_{ij}(g)}{\sum_{g \in \mathcal{N}_j} z_{gj}} \tag{11}$$

where $\mathcal{N}_j$ is the set of nodes from which at least Ω packets have been successfully received by node j (with $\Omega = 2$ in simulations). The network-level PIR is obtained by averaging $\text{E}[\text{PIR}_j]$ over all nodes.

AoI measures the freshness of the most recently received information and applies to a wide range of communication scenarios, including cooperative awareness, perception, and

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coordination. The average AoI for updates from node i to node j is defined as

$$\mathrm{E}[\mathrm{AoI}_{ij}] = \lim_{T\to\infty} \frac{1}{T}\int_0^T A_{ij}(t)\,dt \;\approx\; \frac{\sum_{g=1}^{z_{ij}} \frac{1}{2} Y_{ij}^2(g)}{\sum_{g=1}^{z_{ij}} Y_{ij}(g)} \quad (12)$$

The relation between AoI and PIR can be expressed as

$$\mathrm{E}[\mathrm{AoI}] = \frac{\mathrm{E}[\mathrm{PIR}^2]}{2\cdot \mathrm{E}[\mathrm{PIR}]}, \quad (13)$$

showing that AoI depends on both the mean and the variability of inter-reception intervals. Based on inter-reception intervals, the probability of AoI violation with respect to a threshold A_{th} is defined for each node pair (i, j) as

$$V_{ij}(A_{\text{th}}) = \frac{\sum_{g=2}^{z_{ij}} \max\{0,\, Y_{ij}(g) - A_{\text{th}}\}}{\sum_{g=2}^{m_{ij}} Y_{ij}(g)} \quad (14)$$

The global AoI violation probability is obtained by averaging V_{ij} over node pairs (i, j).

B. Simulation setup

Simulations consider two numerologies, $\mu = 0$ and $\mu = 1$, corresponding to sub-carrier spacings of 15 kHz and 30 kHz, respectively. The only frame parameters kept fixed across numerologies are the bandwidth part, $BWP = 20$ MHz, and the frame duration, $T_f = 10$ ms. The time slot duration depends on the numerology, namely $T_s = 1$ ms for $\mu = 0$ and $T_s = 0.5$ ms for $\mu = 1$. The frame partition into contention time and SCs is adapted accordingly.

For $\mu = 0$, the contention time occupies one slot, corresponding to 1 ms, over the full 20 MHz bandwidth. The remaining nine slots per frame host SCs, i.e., the resource to be allocated. Each SC is composed of 25 Resource Blocks (RBs). With 4 SCs per slot, 36 SCs per frame are available for allocation.

For $\mu = 1$, the full available bandwidth for two slot times is devoted to contention, so that contention time still amounts to 1 ms. The data part of the frame, therefore, consists of the remaining 18 slot times. Assigning still 25 RBs to one SC, each slot accommodates 2 SCs. Then, one frame provides 36 SCs for allocation, as in the case of $\mu = 0$. This design ensures that the two numerologies use the same frame duration, with the same contention time and the same amount of assignable SCs, thus enabling a fair comparison in the subsequent experiments.

Given the numerology dependent SCS and the overall bandwidth, 1200 and 600 sub-carriers are available for $\mu = 0$ and $\mu = 1$, respectively. As we will see in Section IV, reliable sub-carrier detection is attained by using one sub-carrier out of three consecutive ones. Then, 400 and 200 sub-carriers are available for contention in case of $\mu = 0$ and $\mu = 1$, respectively.

Message generation time is set to $T_g = 100$ ms, delay budget to $T_{\text{DB}} = 60$ ms, and AoI violation threshold to three different value: 100 ms, 200 ms and 500 ms. There are 14 Orthogonal Frequency-Division Multiple Access (OFDM) symbols per slot. The channel load coefficient a (see Equation (1)) is used as the independent variable. Simulation time is $T_{\text{sim}} = 50$ s.

As for RCS configuration, the number of contention rounds is fixed to $r = 3$, thus assigning three OFDM symbol times to each round (about 213 µs), with one symbol time gap between subsequent rounds (according to results obtained in Section IV). In contrast to DS and SPS, only 36 SCs are available for allocation per frame, since one 1 ms portion of the frame is reserved for contention. The number of contention groups takes values in the set $\{1, 2, 3, 6, 9, 18, 36\}$, which corresponds to $\{36, 18, 12, 6, 4, 2, 1\}$ SCs associated with each group. Accordingly, the number of sub-carriers per group is $\{400, 200, 133, 66, 44, 22, 11\}$ for $\mu = 0$ and $\{200, 100, 66, 33, 22, 11, 5\}$ for $\mu = 1$.

Table II
SIMULATION PARAMETERS UNDER DIFFERENT NUMEROLOGIES

Unified parameter	$\mu = 0$	$\mu = 1$
Sub-Carrier Spacing (SCS)	15 kHz	30 kHz
Slot duration (T_s)	1 ms	0.5 ms
Total RBs	106	51
Used sub-carriers (M)	400	200
SCs per slot (n_{SC})	4	2
Message generation time (T_g)	100 ms	
Frame duration (T_f)	10 ms	
Bandwidth (BWP)	20 MHz	
RBs per SC	25	
OFDM symbols per slot	14	
Time Delay Budget (T_{DB})	60 ms	
Load coefficient (a)	[0.3; 1.1]	
Simulation time (T_{sim})	50 s	
AoI violation threshold (A_{th})	100, 200, 500 ms	
DS parameter	$\mu = 0$	$\mu = 1$
Number of SCs for allocation in frame (K)	40	
Persistence probability (P_{pers})	0	
RC	1	
SPS parameter	$\mu = 0$	$\mu = 1$
RRI	100 ms	
Number of SCs for allocation in frame (K)	40	
Persistence probability (P_{pers})	0, 0.8	
RC	[5; 15]	
RCS parameter	$\mu = 0$	$\mu = 1$
Max. number of contention rounds (r)	3	
Number of SCs for allocation in frame (K)	36	
Number of contention groups (J)	1, 2, 3, 6, 9, 18, 36	
SCs per contention group (w)	36, 18, 12, 6, 4, 2, 1	
Sub-carriers per group (m_J)	400,200,133, 66,44,22,11	200,100,66, 33,22,11,5

As for SPS configuration, the resource reservation interval is set to $\mathrm{RRI} = 100$ ms, the persistence probability is set to two values $P_{pers} = 0$ and $P_{pers} = 0.8$, and the RC is uniformly drawn from the range $[5, 15]$. In this case, all 40 SCs in each frame are available for allocation. For a more detailed description of the operation of DS and SPS, the reader is referred to [1, 3, 4, 5, 9].

DS and SPS are implemented in accordance with the model presented in [7, 8]. In the SPS implementation, we additionally incorporate the delay budget. T_{DB}. If, during the sensing and resource selection phase, insufficient resources are available for reservation, the packet is considered dropped. This does not occur in the DS case, as transmission is performed in a one-shot manner. Another important modification of the model is that the channel load can exceed one, i.e., $N > K$. It is also assumed that all nodes are within mutual communication range, and that each message requires a single SC for successful transmission, meaning it fits within one transport block.

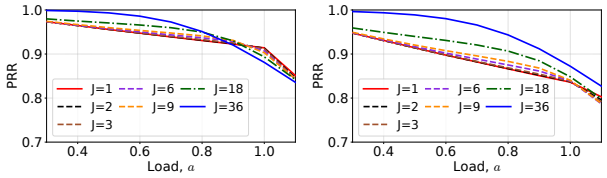
Table II summarizes the value of all relevant parameters.

C. Tuning of RCS parameters

In this section, we compare different settings of RCS parameters, focusing on different numbers of contention groups J under various numerologies. All metrics presented are functions of the load a.

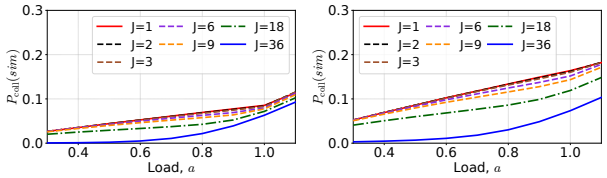
Figure 2 shows the PRR for different values of J. As the load increases, the PRR decreases for both values of

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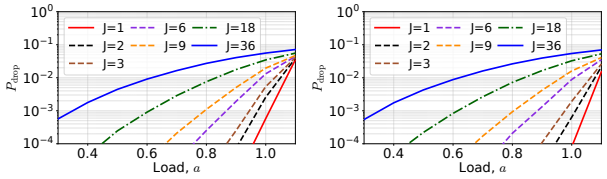
(a) $\mu = 0$ (b) $\mu = 1$

Figure 2. PRR vs. load for different numbers of contention groups J.



(a) $\mu = 0$ (b) $\mu = 1$

Figure 3. $P_{\text{coll}}(sim)$ vs. load for different numbers of contention groups J.



(a) $\mu = 0$ (b) $\mu = 1$

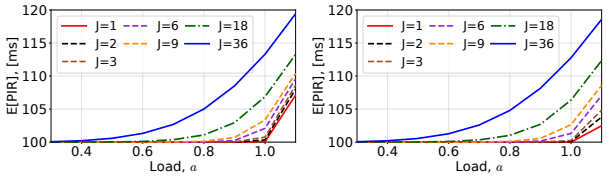
Figure 4. P_{drop} vs. load for different numbers of contention groups J.

μ. For $\mu = 0$, at a load of $a = 0.9$, $J = 36$ is no longer optimal, and $J = 1$ becomes the best configuration in terms of successful packet delivery. In contrast, for $\mu = 1$, $J = 36$ consistently achieves the highest performance, while $J = 1$ yields the lowest. This occurs because the number of sub-carriers within a group for $\mu = 1$ is half that of the corresponding configuration at $\mu = 0$, leading to increased packet loss due to collisions or drops. For all other metrics, $J = 1$ and $J = 36$ represent boundary conditions, with intermediate values of J yielding performance between these extremes.

Figure 3 illustrates the collision probability, i.e., the probability that multiple winners emerge during the contention phase for a given SC, causing simultaneous use of the same resources by multiple nodes. For $\mu = 0$, this probability is lower than for $\mu = 1$. For $J = 36$, performance remains nearly invariant with numerology, while for a smaller number of contention groups under $\mu = 1$, performance deteriorates. The collision probability increases with load in all cases.

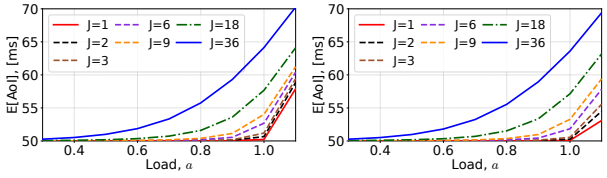
Figure 4 presents the packet drop probability. Here, $J = 36$ exhibits a substantially higher drop probability than $J = 1$. For $J = 1$ under low load, the drop probability is near zero and increases sharply only when $a > 0.9$. The figure uses a logarithmic scale, with maximum values remaining in the range [0.01, 0.1] even at $a = 1$. The drop probability is largely insensitive to numerology. The higher drop probability for $J = 36$ is due to the presence of only a single SC in the group, resulting in more packets being discarded after the T_{DB} during repeated contention, where competition is more intense. Moreover, with $J = 36$ the probability that some group is deserted is not negligible, thus reducing the effectiveness of resource assignment.

Figure 5 and Figure 6 show the mean PIR and mean AoI, respectively. Both metrics increase with load due to more frequent collisions and longer intervals between successful



(a) $\mu = 0$ (b) $\mu = 1$

Figure 5. $E[PIR]$ vs. load for different numbers of contention groups J.



(a) $\mu = 0$ (b) $\mu = 1$

Figure 6. $E[AoI]$ vs. load for different numbers of contention groups J.

transmissions. For $\mu = 0$ and $\mu = 1$, $J = 1$ yields the best performance. The absolute differences in performance remain moderate, on the order of 15 ms under high load.

These results reveal two representative operating regimes for RCS. At low load, $J = 36$ achieves the highest probability of successful reception and the lowest collision probability, as losses are dominated by collisions and a larger number of groups reduces contention effectively. As the load increases, this advantage diminishes, and performance across group configurations converges. In the high-load regime ($a \geq 0.9$), $J = 1$ becomes preferable, offering better timeliness and the highest successful transmission probability, while $J = 36$ becomes slightly less effective. Overall, RCS maintains satisfactory performance, with collision probabilities on the order of 10^{-1} at the maximum considered load. In this regime, packet loss is increasingly influenced by drop events, as packets cannot always be delivered within the available time budget.

Therefore, in Section III-D, we consider only the boundary cases $J = 1$ and $J = 36$ for RCS and restrict the analysis to $\mu = 1$, as DS and SPS performance is independent of μ with given channel configuration, while RCS is evaluated in its worst-case PRR configuration.

D. Comparison with standardized sidelink multiple access algorithms

In this section, we compare the configurations RCS with $J = 1$ and $J = 36$, DS, and the standardized persistence probability values 0 and 0.8 for SPS. We analyze the performance of the standardized resource selection algorithms and the proposed RCS using the PRR, mean PIR, and mean AoI. Additionally, we examine the AoI violation probability for three thresholds: 100 ms, 200 ms, and 500 ms. All metrics presented are functions of the load a.

Figure 7 presents three metrics: PRR (left), mean PIR (center), and mean AoI (right, plotted on a logarithmic scale). The results show that the proposed RCS can outperform the standardized SPS and DS under periodic network traffic. From the perspective of PRR, the advantage of RCS over SPS is moderate, as the curves for different RCS and SPS configurations intersect. Nevertheless, RCS consistently outperforms DS and SPS with a persistence probability of 0 across all load conditions and matches the PRR of SPS only when the persistence probability is 0.8.

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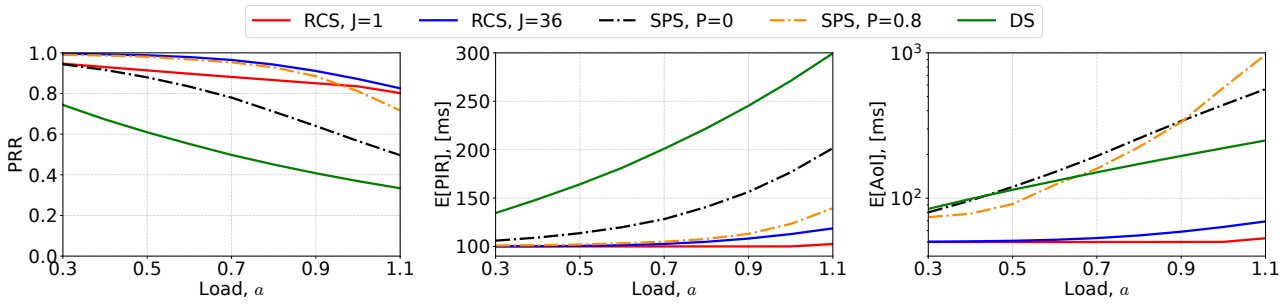


Figure 7. Comparison of DS, SPS, and RCS performance in terms of PRR (left), $\mathbb{E}[\mathrm{PIR}]$ (center), and $\mathbb{E}[\mathrm{AoI}]$ (right).

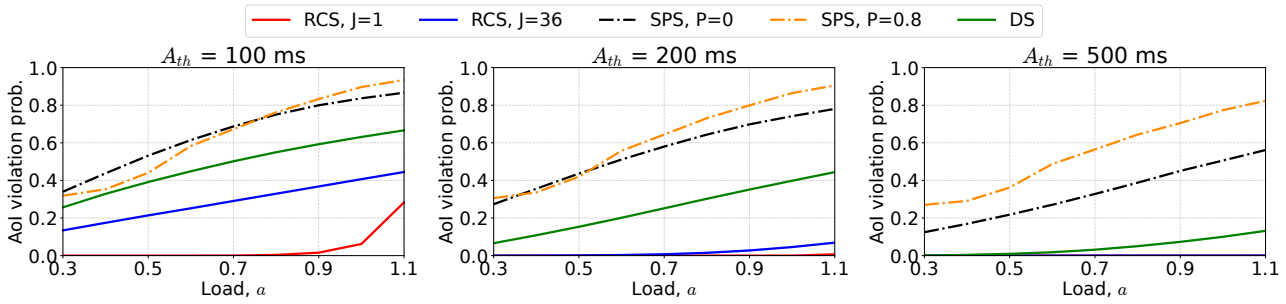


Figure 8. Comparison of DS, SPS, and RCS in terms of AoI violation probability for different thresholds A_{th}: left ($A_{th} = 100$ms), center ($A_{th} = 200$ms), and right ($A_{th} = 500$ms).

The analysis of the mean PIR reveals a clear trend: as the load increases, the average interval between successfully delivered packets grows. This increase is steeper for DS and SPS, particularly at high channel loads around 0.8–0.9. At low loads, SPS with a persistence configuration of 0.8 is comparable to RCS, but under higher loads, RCS clearly outperforms SPS.

Considering the mean AoI as a function of load, the proposed RCS significantly surpasses all standardized resource selection algorithms. In DS, collisions are frequent but random and uncorrelated, whereas SPS experiences consecutive persistent collisions and burst losses, leading to a sharper increase in mean AoI.

This behavior is further illustrated in Figure 8, which shows the AoI violation probability. RCS outperforms SPS and DS across all thresholds, including the stringent 100 ms threshold, demonstrating that RCS effectively mitigates persistent collisions and bursty losses.

For SPS, the violation probability changes little across thresholds, indicating that many AoI samples exceed 500 ms, reflecting persistent collisions that block communication for extended periods. In contrast, for DS and RCS, the violation probability decreases with increasing thresholds, indicating fewer extreme AoI values, consistent with observations in [9, 27]. This demonstrates more regular and reliable packet delivery. Notably, for the 200 ms threshold, RCS exhibits extremely low violation probabilities, not exceeding 0.1 even under high load. For the 500 ms threshold, the probability approaches zero, showing that messages are delivered consistently even in the presence of collisions and drops, ensuring stable communication.

These results indicate that RCS outperforms the standardized DS and SPS under the worst-case configuration ($\mu = 1$), providing high packet delivery reliability and regular, stable updates between nodes. This improvement arises from the complete elimination of the sensing and resource reservation procedure, which significantly reduces the probability and duration of persistent collisions.

The obtained results also indicate that a reliability metric, such as PRR, for RCS is fully correlated with timeliness metrics, such as AoI. In contrast, for SPS, this correlation is nontrivial and not immediately evident, due to the hidden issue of persistent collisions, including those that may last for extended periods.

IV. Experimental Proof-of-Concept

This section presents the experimental validation of the RCS scheme. Section IV-A defines metrics for reliability and feasibility. Section IV-B details the hardware, PHY, and synchronization setup. Section IV-C examines key PHY and protocol parameters, and Section IV-D compares experimental results with simulations, highlighting real-world non-idealities.

A. Key performance metrics for experimental evaluation of RCS

The considered reliability metrics are: probability of success (P_{succ}), collision probability ($P_{\mathrm{coll}}(exp)$), no-winner probability (P_{nowin}). Unlike simulations, the experiment yields three possible outcomes: (i) success with a single winner; (ii) collision with multiple winners; and (iii) contention terminating without any winner. Hence it is $P_{\mathrm{succ}}+P_{\mathrm{coll}}(exp)+P_{\mathrm{nowin}} = 1$ and the probability of packet loss is $P_{\mathrm{loss}} = P_{\mathrm{coll}}(exp) + P_{\mathrm{nowin}}$.

In the following calculations, P_{succ} is equivalent to the PRR (Equation (6)), while the collision probability defined in simulations, $P_{\mathrm{coll}}(\mathrm{sim})$ (Equation (7)), corresponds directly to its experimental counterpart, $P_{\mathrm{coll}}(\mathrm{exp})$.

The no-winner probability, observed only in experiments, is defined as the fraction of contentions that end without selecting any winner, relative to the total number of contentions. It is given by:

$$P_{\mathrm{nowin}} = \frac{N_{\mathrm{nowin}}}{N_{\mathrm{total}}} \tag{15}$$

To identify a suitable configuration, we evaluated the false positive probability (P_{fp}), false alarm probability (P_{fa}), and

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Table III
EXPERIMENTAL PARAMETERS

Parameter	Value
Central frequency (f_c)	5 GHz
FFT size (N_{FFT})	32 samples
Numerology (μ)	0
SCS	15 kHz
Slot duration (T_s)	1 ms
Bandwidth (BWP)	480 kHz
OFDM symbols per slot	14
Symbol duration	71.428 ms
Cycle Prefix length	3 samples

the detection probability (P_d). A false alarm occurs when a node detects a non-existent tone, potentially causing all nodes to withdraw and leaving the resource unassigned. A missed detection occurs when a transmitted tone is not detected, which may result in multiple nodes claiming the resource and causing a collision. The false positive probability is defined as the probability that a node detects a tone on a sub-carrier that immediately precedes the transmitted one, typically due to spectral leakage caused by the signal's shape.

B. Experimental setup

The experimental setup implements a wireless node prototype capable of simultaneous tone transmission and channel sensing, as required for frequency-domain RCS operation [43]. The system is built on a Software Defined Radio (SDR) platform based on Ettus USRP B200 devices, enabling flexible physical layer control. Signal processing and protocol logic are implemented in GNU Radio, allowing modular integration of the contention mechanism with the radio front-end. Synchronization across nodes is achieved via an Ettus CDA-2990 OctoClock, which provides a common frequency reference and aligned sampling time.

The SDR implementation introduces practical impairments, including noise, interference, limited dynamic range, and oscillator instability. These effects require careful parameter tuning, particularly for guard band allocation and sub-carrier selection, to ensure reliable operation under FR1 sidelink specifications [45]. The adopted PHY configuration, summarized in Table III, balances standard compliance with real-time processing constraints. The system operates with numerology $\mu = 0$, corresponding to a SCS of 15 kHz and a slot duration of 1 ms.

Following the RCS principle, contention levels are mapped onto OFDM sub-carriers. Due to computational constraints, the FFT size is fixed to $N_{FFT} = 32$, from which a subset of 10 sub-carriers is selected to represent contention levels.

Each node transmits a single OFDM tone by activating one randomly selected sub-carrier while sensing the channel simultaneously. If a tone is detected at a lower frequency, the node withdraws; otherwise, it proceeds to the next contention round. A tone is considered valid if its power exceeds a threshold determined by the noise floor.

To emulate a contention with multiple stations, one of the available USRP devices was programmed to represent more than a single node, allowing the total number of competing nodes n to range between 3 and 10. At each round, it generates simultaneous OFDM tones over m sub-carriers and executes the RCS procedure. The evaluation is based on an extensive experimental campaign comprising more than 10^5 contention cycles per configuration.

C. Experimental parameters tuning

The sub-carrier selection (see Table IV) mitigates hardware non-idealities by introducing a spacing of 2 sub-carriers between contention levels, reducing inter-carrier interference, and excluding the DC and mirror frequencies to avoid sensing errors due to local oscillator leakage and image components. This spacing also limits false positives caused by spectral leakage into adjacent bins, which may lead nodes to prematurely withdraw from contention. Evaluation of the false positive probability for different guard spacings shows that using 2 guard sub-carriers achieves P_{fp} of approximately 10^{-2} (see Figure 9a), which is acceptable for the experimental setup.

The round duration determines the trade-off between detection reliability and the number of rounds per slot: short durations degrade detection performance, while longer durations reduce the number of available rounds. Given the 14-symbol slot structure, with the first and last symbols reserved for PHY functions [1] and a guard symbol required between consecutive rounds, four time-slot configurations are feasible:

- 6 rounds of 1 symbol each, with 6 guard symbols;
- 4 rounds of 2 symbols each, with 4 guard symbols;
- 3 rounds of 3 symbols each, with 3 guard symbols;
- 2 rounds of 5 symbols each, with 2 guard symbols.

The false alarm (P_{fa}) and detection (P_d) probabilities were evaluated for each round duration (see Figures 9b and 9c). A duration of three OFDM symbols achieves acceptable performance ($P_{fa} \approx 10^{-4}$, $P_d \approx 0.999$), whereas shorter durations lead to significantly higher error rates.

Specifically, one-symbol rounds provide insufficient energy integration, resulting in poor detection performance ($P_d \approx 0.5$). Two-symbol rounds improve detection but exhibit elevated false alarm rates due to limited robustness against noise and spectral leakage. A duration of three OFDM symbols provides the best trade-off, ensuring reliable detection while allowing multiple contention rounds within a single slot. This configuration is adopted as the baseline for the hardware implementation of the RCS algorithm.

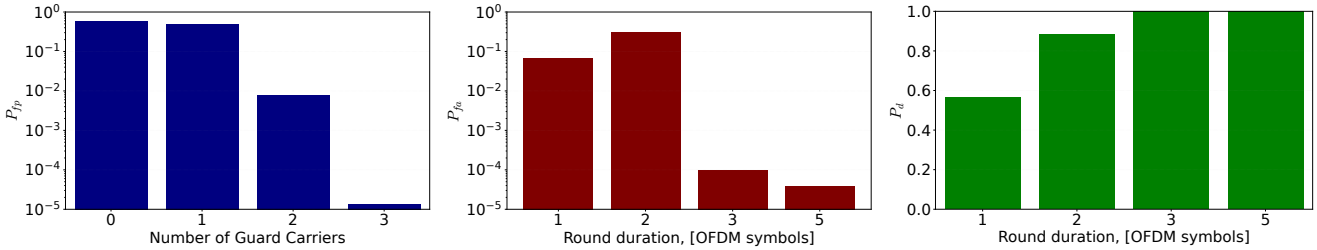
D. Experimental results

The experimental success probability, compared with simulation results, is shown in Figure 10. Please note that the simulation parameters shown in Figure 10 are consistent with the experimental parameters reported in Table III. The comparative analysis reveals that the experimental P_{succ} measured on the USRP nodes is slightly lower than the values obtained through simulations. This performance gap is directly attributable to the emergence of P_{nowin} within the hardware based setup. While the simulation environment assumes ideal conditions where every non-colliding contention results in a success, the physical implementation introduces real-world impairments. Specifically, a small fraction of these

Table IV
MAPPING OF CONTENTION SUB-CARRIERS INDICES AND FREQUENCY OFFSETS WITH RESPECT TO THE CENTER FREQUENCY.

Index	Offset (kHz)	Index	Offset (kHz)
-15	-225	+1	+15
-12	-180	+4	+60
-9	-135	+7	+105
-6	-90	+10	+150
-3	-45	+13	+195

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(a) P_{fp} as a function of guard carriers number (b) P_{fa} as a function of round duration (c) P_d as a function of round duration

Figure 9. Results of experimental parameters tuning.

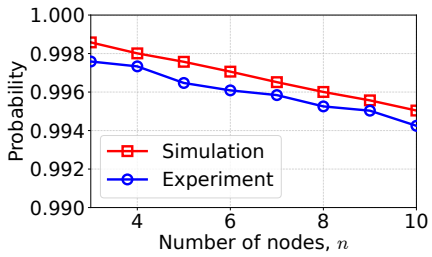


Figure 10. Comparison between experimental and simulation-based success probability.

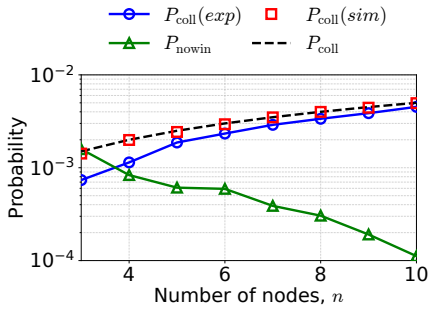


Figure 11. Comparison of experimental and simulation-based collision probabilities, the theoretical upper bound on collision probability, and the no-winner probability.

contention slots is lost due to false alarms during the sensing phase. In such instances, environmental impulse noise or interference is erroneously interpreted as an active tone from a competitor, leading nodes to defer and resulting in an inconclusive contention cycle even in the absence of actual collisions.

The no-winner probability is illustrated in Figure 11, which compares simulation and experimental results. The figure reports the theoretical upper bound on collision probability from Equation (5) (black curve), the simulated collision probability (red markers), and the experimentally measured collision (blue curve) and no-winner (green curve) probabilities.

First, the presence of no-winner events (P_{nowin}) reduces the number of contention phases that can result in collisions, as some rounds terminate prematurely due to sensing errors. Second, physical-layer impairments may induce a collision-to-success transition. When two nodes select the same sub-carrier in the final round, a collision is expected; however, sensing errors or noise fluctuations may cause one node to detect a spurious lower-frequency tone and withdraw. As a result, the other node gains access to the resource, effectively converting a potential collision into a successful transmission. Although infrequent, this effect further reduces the observed P_{coll} compared to the ideal model.

The experimental results indicate that the no-winner contention probability (P_{nowin}) decreases as the number of contending nodes increases. This trend can be attributed to the fact that a higher density of participants increases the likelihood that false detections caused by noise are masked by the actual tones transmitted by the stations. In a scenario with a limited number of available sub-carriers, the event in which a station erroneously withdraws due to a low index false tone becomes increasingly rare as the spectrum becomes more populated. Conversely, the collision probability (P_{coll}) scales proportionally with the number of contending nodes, as the statistical likelihood of multiple stations selecting an identical sub-carrier in the final round increases. This inverse relationship between P_{nowin} and P_{coll} is depicted in Figure 11, which shows how the growth of n shifts the system's primary impairment from sensing-induced deferrals to physical collisions. The experimental data reveal a clear transition in the dominant impairment factor as the network density increases.

V. Conclusion

This paper introduced RCS, a novel resource allocation algorithm for NR-V2X and towards 6G sidelink communications that departs from DS and SPS. By replacing reservation-based mechanisms with a multi-round contention process, RCS enables fully distributed, feedback-driven resource selection, mitigating persistent collisions and adapting to dynamic vehicular environments. Simulation results demonstrate that RCS consistently outperforms DS and SPS in terms of reliability and timeliness, achieving higher PRR, lower loss rates, and improved PIR and AoI, particularly under high load conditions. The feasibility of RCS is further validated through an SDR-based proof-of-concept. Despite hardware impairments, experimental results align with theoretical and simulation trends, confirming that RCS operates reliably in practical settings and represents a viable solution for decentralized resource allocation in 5G NR and beyond.

Future work will focus on detailed modeling of the proposed RCS algorithm under more realistic conditions, including aperiodic traffic, hidden nodes, signal propagation and fading models, as well as experimental investigations of scenarios involving multiple winners in the contention phase.

Acknowledgments

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References

[1] M. H. C. Garcia et al., "A tutorial on 5G NR V2X communications," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 3, pp. 1972–2026, 2021.

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[2] Z. Ali, S. Lagén, L. Giupponi, and R. Rouil, "3GPP NR V2X mode 2: overview, models and system-level evaluation," *IEEE Access*, vol. 9, pp. 89 554–89 579, 2021.

[3] 3GPP, "NR; Medium Access Control (MAC) protocol specification (Release 19)," 3GPP, TS 38.321 V19.2.0, Dec. 2025.

[4] 3GPP, "Overall description of Radio Access Network (RAN) aspects for Vehicle-to-everything (V2X) based on LTE and NR (Release 19)," 3GPP, TR 37.985 V19.0.0, Oct. 2025.

[5] L. Lusvarghi, A. Molina-Galan, B. Coll-Perales, J. Gozalvez, and M. L. Merani, "A Comparative Analysis of the Semi-Persistent and Dynamic Scheduling Schemes in NR-V2X Mode 2," *Vehicular Communications*, p. 100 628, 2023.

[6] A. Bazzi, C. Campolo, A. Molinaro, A. O. Berthet, B. M. Masini, and A. Zanella, "On Wireless Blind Spots in the C-V2X Sidelink," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 8, pp. 9239–9243, 2020.

[7] A. Rolich, I. Turcanu, A. Vinel, and A. Baiocchi, "Understanding the impact of persistence and propagation on the Age of Information of broadcast traffic in 5G NR-V2X sidelink communications," *Computer Networks*, vol. 248, p. 110 503, 2024.

[8] A. Rolich, I. Turcanu, A. Vinel, and A. Baiocchi, "Impact of Persistence on the Age of Information in 5G NR-V2X Sidelink Communications," in *2023 21st Mediterranean Communication and Computer Networking Conference (MedComNet)*, 2023, pp. 15–24.

[9] A. Rolich, M. Yildiz, I. Turcanu, A. Vinel, and A. Baiocchi, "Rethinking Persistent Scheduling in 5G New Radio Vehicle-to-Everything Sidelink Communications," *IEEE Access*, vol. 13, pp. 164 065–164 083, 2025.

[10] S. Ha, W. Yoo, H. Kim, and J.-M. Chung, "C-V2X Adaptive Short-Term Sensing Scheme for Enhanced DENM and CAM Communication," *IEEE Wireless Communications Letters*, vol. 11, no. 3, pp. 593–597, 2022.

[11] V. Todisco, Z. Wu, and A. Bazzi, "Improving NR-V2X Autonomous Mode Through Resource Re-Evaluation," in *2024 IEEE Vehicular Networking Conference (VNC)*, 2024, pp. 125–131.

[12] A. Molina-Galan, L. Lusvarghi, B. Coll-Perales, J. Gozalvez, and M. L. Merani, "On the Impact of Re-Evaluation in 5G NR V2X Mode 2," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 2, pp. 2669–2683, 2024.

[13] A. Molina-Galan, J. Gozalvez, and B. Coll-Perales, "A Selective Re-Evaluation Mechanism for 5G NR V2X Mode 2 Communications," *IEEE Transactions on Vehicular Technology*, vol. 75, no. 2, pp. 2939–2951, 2026.

[14] Q. Han, M. Zhou, L. Zeng, L. Ye, and L. Yang, "NR-V2X Mode2 Resource Selection Window Size Adjustment Based on Resource Priority Index," in *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, 2024, pp. 411–416.

[15] M. Shehata, C. Ciochina, and J.-C. Sibel, "On the Performance of Inter User Coordination for Resource Allocation Enhancement in NR V2X Mode 2," in *Communication Technologies for Vehicles*, Cham: Springer International Publishing, 2021, pp. 3–14.

[16] S. An and K. Chang, "Enhancing Reliability in 5G NR V2V Communications Through Priority-Based Groupcasting and IR-HARQ," *IEEE Access*, vol. 11, pp. 72 717–72 731, 2023.

[17] C. Mahabal and T. Shimizu, "Impact of HARQ Retransmissions on Information Age in 5G NR Sidelink," in *2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall)*, 2024, pp. 1–6.

[18] Q. Fu, J. Liu, J. Wang, and J. Li, "SecHARQ: A Secure Scheme for HARQ-Assisted NR V2X Communications," *IEEE Transactions on Network and Service Management*, vol. 22, no. 2, pp. 2199–2213, 2025.

[19] M. Parvini, P. Schulz, and G. Fettweis, "Resource Allocation in V2X Networks: From Classical Optimization to Machine Learning-Based Solutions," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 1958–1974, 2024.

[20] M. A. Ali, S. A. Khan, S. Aldirmaz Colak, S. Kosunalp, and T. Iliev, "Towards 6G C-V2X Networks: A Comprehensive Survey on Mobility Management, Multi-RAT Coexistence, and Machine Learning (3M) Framework for C-ITS," *Electronics*, vol. 15, no. 5, 2026.

[21] A. Hegde, R. Song, and A. Festag, "Radio Resource Allocation in 5G-NR V2X: A Multi-Agent Actor-Critic Based Approach," *IEEE Access*, vol. 11, pp. 87 225–87 244, 2023.

[22] Z. Li, P. Wang, Y. Shen, and S. Li, "Reinforcement Learning-Based Resource Allocation Scheme of NR-V2X Sidelink for Joint Communication and Sensing," *Sensors*, vol. 25, no. 2, 2025.

[23] M. Montaño, J. Gomez-Ponce, M. Antonieta-Alvarez, F. Novillo, and R. Cajo, "Adaptive MCS Optimization with Feature Selection and Machine Learning for C-V2X Sidelink," in *Advanced Research in Technologies, Information, Innovation and Sustainability*, Springer Nature, 2026, pp. 311–323.

[24] M. M. Saad, M. A. Tariq, M. Ajmal, D. Kim, and G. Srivastava, "Federated Multiagent Reinforcement Learning for Resource Allocation in NR-V2X Mode 2," *IEEE Internet of Things Journal*, vol. 12, no. 13, pp. 23 402–23 417, 2025.

[25] T. Xiaolu, S. Yan, X. Yaqi, C. Shanzhi, and G. Yuming, "AoI and TTC based resource allocation in C-V2X sidelink via multi-agent reinforcement learning," *China Communications*, vol. 22, no. 8, pp. 281–297, 2025.

[26] A. Rolich, M. Yildiz, and A. Baiocchi, "Safety-Critical Delays in Vehicular Networks Toward 6G: A Novel Metric for Disconnection Time Assessment," *IEEE Communications Letters*, vol. 30, pp. 1860–1864, 2026.

[27] A. Rolich, M. Yildiz, I. Turcanu, A. Vinel, and A. Baiocchi, "On the Trade-off Between AoI Performance and Resource Reuse Efficiency in 5G NR V2X Sidelink," in *IEEE Vehicular Networking Conference (VNC)*, Porto, Portugal: IEEE, Jun. 2025.

[28] W.-D. Shen and H.-Y. Wei, "Age-of-Information Performance Analysis in Power-Efficient Sidelink Communications," *IEEE Internet of Things Journal*, vol. 12, no. 20, pp. 42 116–42 132, 2025.

[29] F. Peng, Z. Jiang, S. Zhang, and S. Xu, "Age of Information Optimized MAC in V2X Sidelink via Piggyback-Based Collaboration," *IEEE Transactions on Wireless Communications*, vol. 20, no. 1, pp. 607–622, 2021.

[30] L. Cao, H. Yin, R. Wei, and L. Zhang, "Optimize Semi-Persistent Scheduling in NR-V2X: An Age-of-Information Perspective," in *2022 IEEE Wireless Communications and Networking Conference (WCNC)*, 2022, pp. 2053–2058.

[31] A. Rolich, I. Turcanu, and A. Baiocchi, "AoI-Aware and Persistence-Driven Congestion Control in 5G NR - V2X Sidelink Communications," in *2024 22nd Mediterranean Communication and Computer Networking Conference (MedComNet)*, 2024, pp. 1–4.

[32] A. Rolich, M. Yildiz, I. Turcanu, A. Vinel, and A. Baiocchi, "From Latency to Value of Information: A Review of Timeliness Metrics for Safe Transportation Systems in the 6G Era," *IEEE Vehicular Technology Magazine*, pp. 2–11, 2026.

[33] A. Bazzi et al., "Toward 6G Vehicle-to-Everything Sidelink: Nonorthogonal Multiple Access in the Autonomous Mode," *IEEE Vehicular Technology Magazine*, vol. 18, no. 2, pp. 50–59, 2023.

[34] V. Todisco, C. Campolo, A. Molinaro, A. O. Berthet, R. A. Stirling-Gallacher, and A. Bazzi, "On the Performance of SIC-based NOMA in the C-V2X Sidelink Autonomous Mode," in *2023 IEEE Conference on Standards for Communications and Networking (CSCN)*, 2023, pp. 66–72.

[35] T. Hirai, T. Kimura, and N. Wakamiya, "Spatial Performance Analysis of Autonomous Sidelink Cellular-V2X with NOMA," in *2022 IEEE Global Communications Conference (GLOBECOM)*, 2022, pp. 1205–1210.

[36] T. Fuchikami, T. Hirai, and N. Wakamiya, "Spatial Performance Analysis of V2X with Sidelink NOMA with Multiple Interfering Vehicles," in *2024 International Conference on Consumer Electronics - Taiwan (ICCE-Taiwan)*, 2024, pp. 605–606.

[37] C. Shin, E. Farag, H. Ryu, M. Zhou, and Y. Kim, "Vehicle-to-Everything (V2X) Evolution From 4G to 5G in 3GPP: Focusing on Resource Allocation Aspects," *IEEE Access*, vol. 11, pp. 18 689–18 703, 2023.

[38] A. N. Al-Najjar, M. F. A. Rasid, F. Hashim, F. A. Ahmad, and A. Jamalipour, "A systematic literature review in distributed resource allocation for C-V2X," *Ingénierie des Systèmes d'Information*, vol. 29, no. 3, pp. 771–808, 2024.

[39] Annu and P. Rajalakshmi, "Towards 6G V2X Sidelink: Survey of Resource Allocation—Mathematical Formulations, Challenges, and Proposed Solutions," *IEEE Open Journal of Vehicular Technology*, vol. 5, pp. 344–383, 2024.

[40] C. Bin Ali Wael, E. Hadj Dogheche, N. Armi, A. Subekti, and I. Dayoub, "Leveraging 3GPP Features and Optimization Techniques for 5G NR-V2X Resource Allocation: A Survey," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 6, pp. 967–994, 2025.

[41] W. Zame, J. Xu, and M. van der Schaar, "Winning the Lottery: Learning Perfect Coordination With Minimal Feedback," *IEEE Journal of Selected Topics in Signal Processing*, vol. 7, no. 5, pp. 846–857, 2013.

[42] A. Baiocchi, I. Tinnirello, D. Garlisi, and A. L. Valvo, "Random access with repeated contentions for emerging wireless technologies," in *IEEE INFOCOM 2017 - IEEE Conference on Computer Communications*, 2017, pp. 1–9.

[43] A. Baiocchi, D. Garlisi, A. L. Valvo, G. Santaromita, and I. Tinnirello, "'Good to repeat': Making random access near-optimal with repeated contentions," *IEEE Transactions on Wireless Communications*, vol. 19, no. 1, pp. 712–726, 2019.

[44] 3GPP, "Study on evaluation methodology of new Vehicle-to-Everything (V2X) use cases for LTE and NR; (Release 15)," 3GPP, TR 37.885 V15.3.0, Jun. 2019.

[45] 3GPP, "NR; Physical channels and modulation (Release 19)," 3GPP, TS 38.311 V19.3.0, Mar. 2026.

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